

Magic: An Interactive VLSI Layout Tool

Magic layout tool has the capability to incorporate expertise about design rules and connectivity directly into a layout system in order to implement some powerful interactive operations

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Magic is an easy-to-use VLSI layout tool that has been popular with universities and small companies. It can be seen as an open source EDA alternative to the various commercial EDA tools out there. It provides many useful features for streamlining your product design flow.

What's noteworthy about Magic

Magic uses simplified design rules and circuit structures that make designing easier, but produces circuits that are of lower density than what you

would get with more complex rules and structures. This density sacrifice is, however, compensated by the reduced design time. Magic permits only Manhattan designs, that is, where edges are either vertical or horizontal.

Magic combines circuit knowledge with an interactive editor interface to speed up your manufacturing design processes. It allows you to enter and modify your designs quickly. The built-in knowledge of layout rules, circuits and transistors, and the knowledge of efficient routing, helps you re-arrange your circuit as a circuit, and not just a collection of geometrical objects. This makes any modification process more efficient and less com-

plex. Most importantly, it allows you to focus on your design while the software takes care of mundane tasks like following layout rules.

Now let us have a look at some key features that set it apart from the other layout editors available.

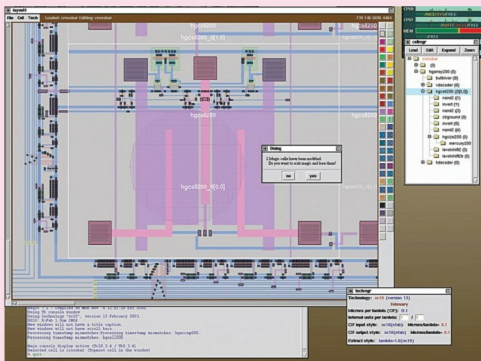
Efficient corner-stitching technique

Like most layout editors, Magic also uses cells. However, the contents of these cells are represented using corner-stitching technique, which provides efficient searching options and allows for quick modification of the database.

Each point in the x-y plane is contained in exactly one tile and empty space is represented explicitly. These tiles are linked together at their corners. This allows for fast and efficient algorithms for searching, creation or deletion. Though it requires more storage space, this technique allows for efficient two-dimensional searches and permits fast updates.

Interactive interface

Magic is usable through an interface that features colour graphics and also allows the use of a mouse to design basic cells. These basic cells can be hierarchically combined to make larger structures. The tool also provides multiple overlapping windows that can refer to different portions of a single cell or to completely



Magic screenshot